

SYSTEM DESIGN CONCEPTS

EVERY ENGINEER SHOULD KNOW



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SYSTEM DESIGN FUNDAMENTALS

Scalability · Latency vs Throughput · Availability & SLO · Idempotency · REST · GraphQL · gRPC · WebSockets · Sync vs Async

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TRAFFIC, SCALING & CACHING

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DATA & STORAGE

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AI SYSTEM ARCHITECTURE · PART 1 OF 2

Vector Database · Embedding Model · RAG · Chunking · Hybrid Search · Semantic Cache · Prompt Engineering · Function Calling · MCP

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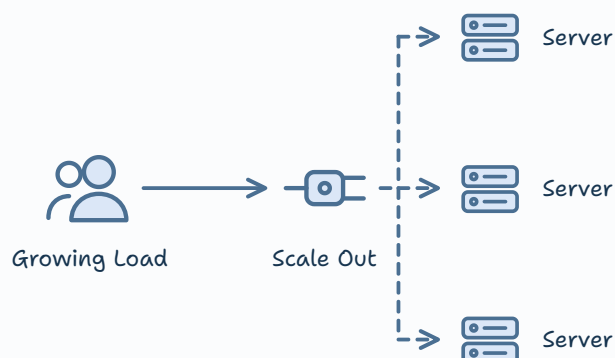
AI SYSTEM ARCHITECTURE · PART 2 OF 2

AI Agent · Multi-Agent Systems · Guardrails · LLM Gateway · Fine-Tuning vs RAG · Inference Optimisation · Evals & LLM-as-Judge · LLM Observability · Human in the Loop

> SYSTEM DESIGN FUNDAMENTALS

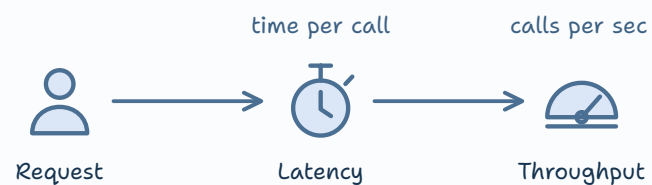
EVERY ENGINEER SHOULD KNOW

1 SCALABILITY



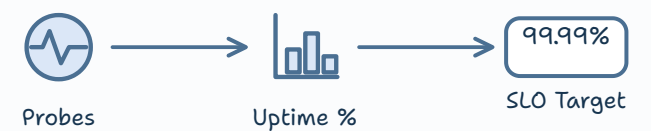
Absorb more load by adding machines (horizontal) or power (vertical).

2 LATENCY VS THROUGHPUT



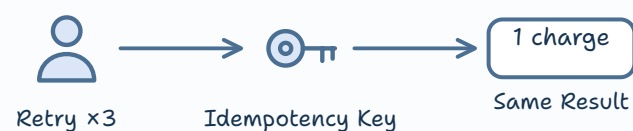
Latency is time for one request. Throughput is requests finished per second.

3 AVAILABILITY & SLO



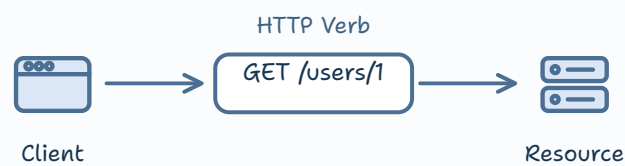
The share of time a system answers correctly — measured, targeted, promised.

4 IDEMPOTENCY



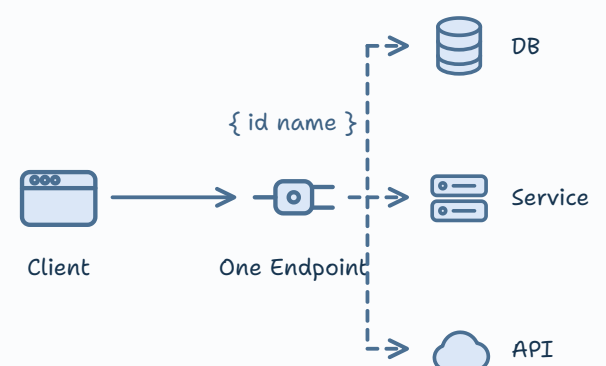
The same request sent twice has one effect — which makes retries safe.

5 REST



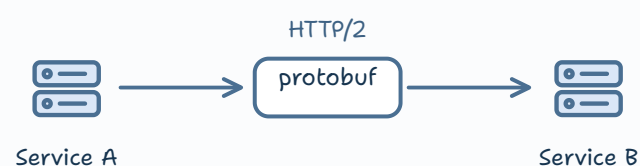
Stateless resources addressed by URL and manipulated with standard HTTP verbs.

6 GRAPHQL



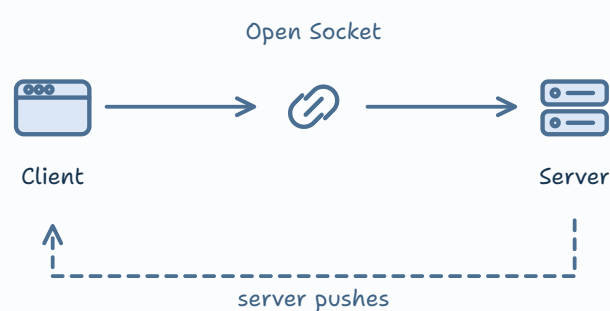
One endpoint where clients ask for exactly the fields they need — no over-fetching.

7 GRPC



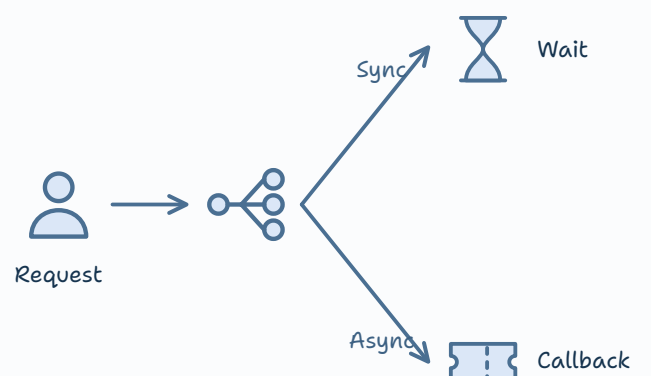
Fast binary RPC over HTTP/2 with typed contracts — ideal for service-to-service calls.

8 WEBSOCKETS



One long-lived two-way connection for real-time push instead of constant polling.

9 SYNC VS ASYNC

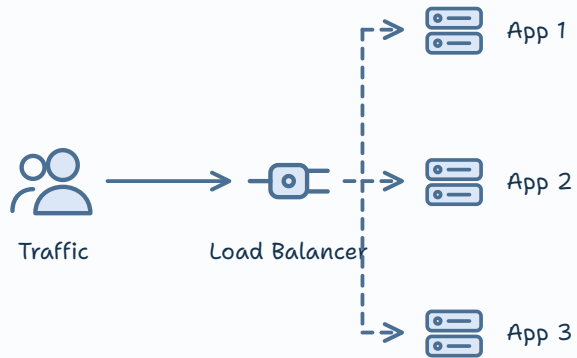


Block and wait for the answer, or take a ticket and get called back later.

> TRAFFIC, SCALING & CACHING

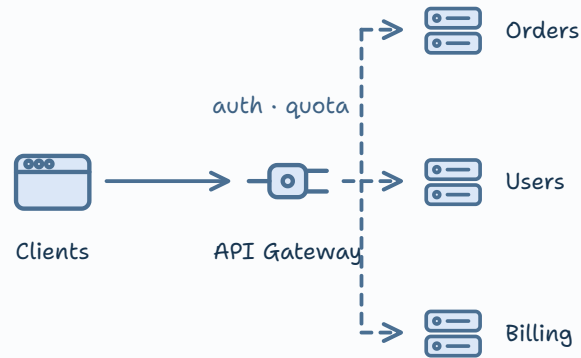
SYSTEM DESIGN CONCEPTS

10 LOAD BALANCING



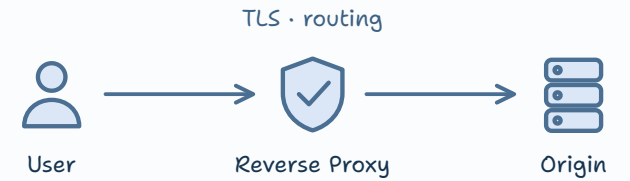
Spread incoming traffic across healthy servers so no single one drowns.

11 API GATEWAY



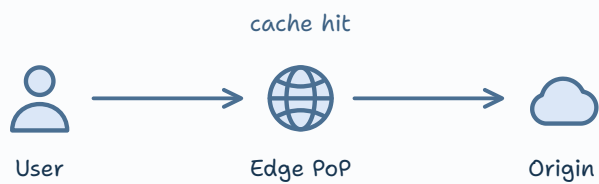
One front door handling auth, routing, quotas and aggregation for many services.

12 REVERSE PROXY



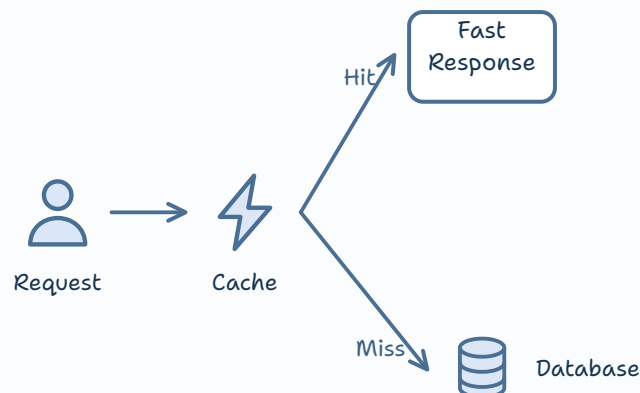
Sits in front of origins to terminate TLS, route, compress and hide topology.

13 CDN



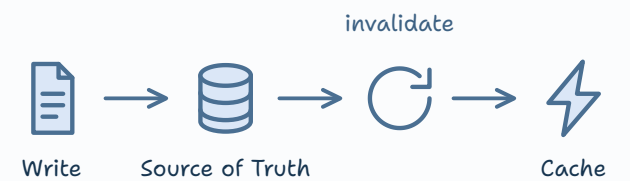
Cache static bytes at edge locations so users are served from a nearby city.

14 CACHING



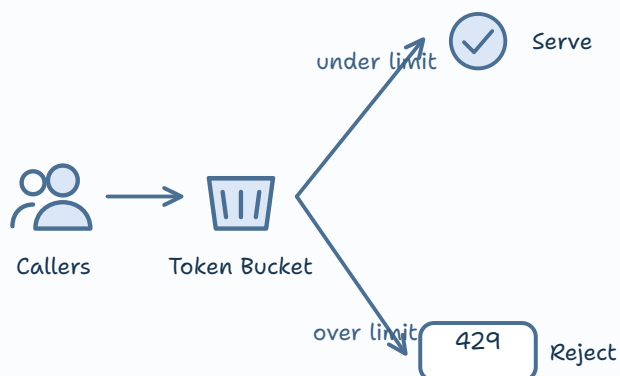
Keep hot answers close to the caller and skip the slow path entirely.

15 CACHE INVALIDATION



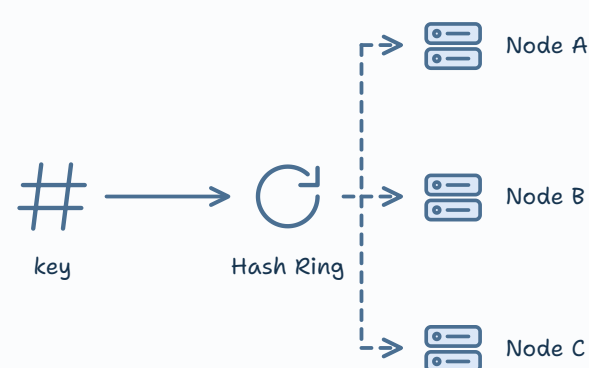
Evict or refresh stale entries — via TTL, write-through or explicit purge.

16 RATE LIMITING



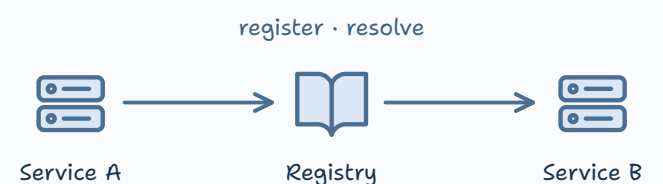
Cap how fast each caller may hit you so one client cannot sink the fleet.

17 CONSISTENT HASHING



Map keys onto a ring so adding a node reshuffles only a small slice of them.

18 SERVICE DISCOVERY



Services look each other up in a registry as instances start, move and die.

19 SQL VS NOSQL



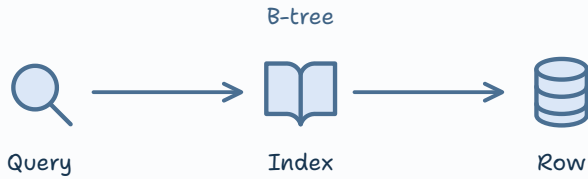
SQL · joins



NoSQL · scale

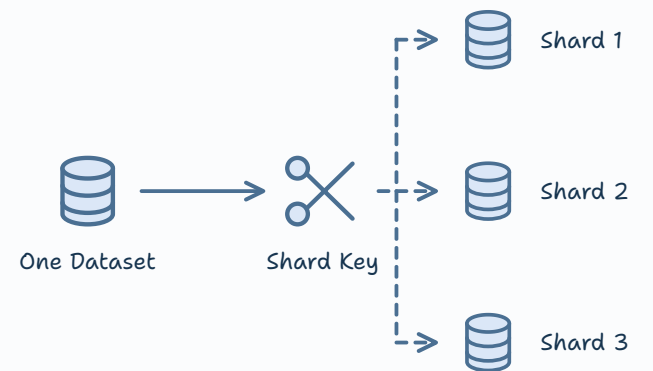
Joins, schema and transactions versus horizontal scale and flexible shapes.

20 INDEXING



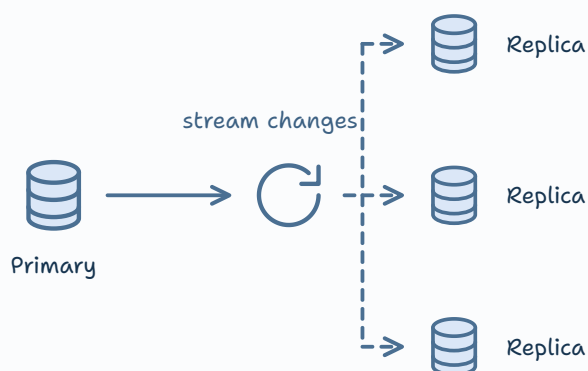
Pay a little on every write so reads stop scanning the whole table.

21 SHARDING



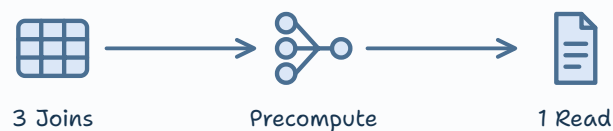
Split one dataset across machines by a shard key so each holds a slice.

22 REPLICATION



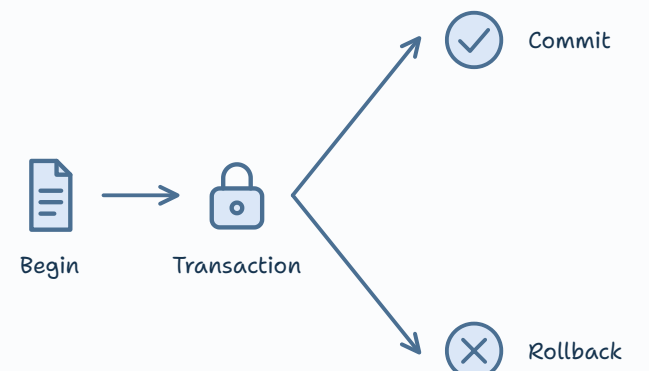
Keep copies of the data so reads scale out and a dead node is survivable.

23 DENORMALIZATION



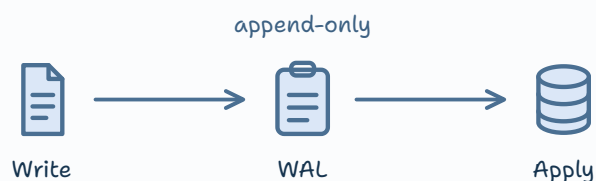
Duplicate data on purpose to turn several expensive joins into one read.

24 ACID TRANSACTIONS



All of the work commits or none of it does — no half-applied state, ever.

25 WRITE-AHEAD LOG



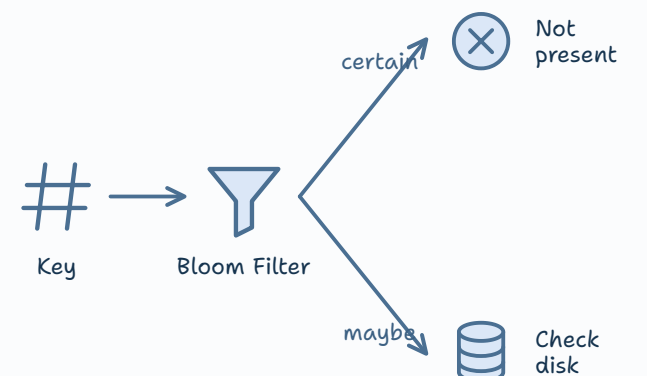
Record the intent durably first, then apply it — replay the log after a crash.

26 OBJECT STORAGE



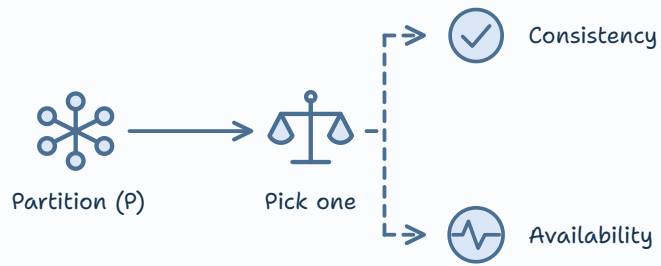
Cheap, effectively endless buckets for blobs, addressed by key over HTTP.

27 BLOOM FILTER



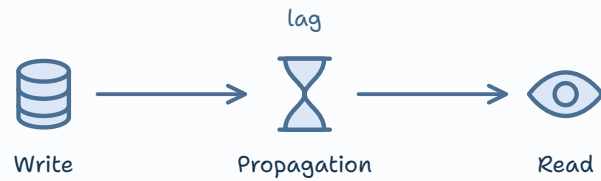
A few kilobytes that answer "definitely not here" and skip a disk lookup.

28 CAP THEOREM



When the network partitions you must choose: stay consistent, or stay available.

29 CONSISTENCY MODELS



How stale a read may be — strong, causal or eventual, each with a price.

30 QUORUM



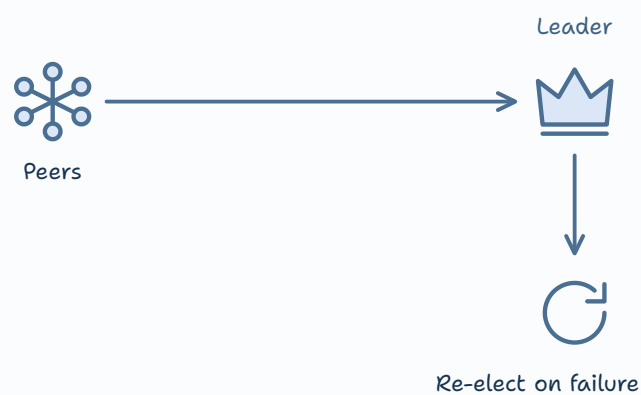
Enough replicas must acknowledge before a write counts: $W + R > N$.

31 CONSENSUS



Raft or Paxos gets independent nodes to agree on one ordered version of truth.

32 LEADER ELECTION



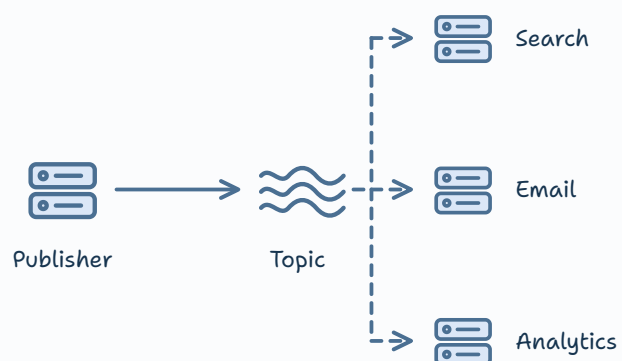
One node is chosen to decide; when it dies the rest elect a replacement.

33 MESSAGE QUEUE



A durable buffer that lets producers outrun consumers without dropping work.

34 PUB / SUB



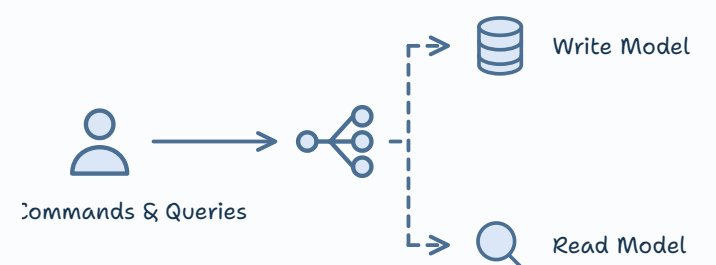
Publish an event once; every interested subscriber gets its own copy.

35 EVENT SOURCING



Store the events, not the current row — replay them to rebuild any state.

36 CQRS



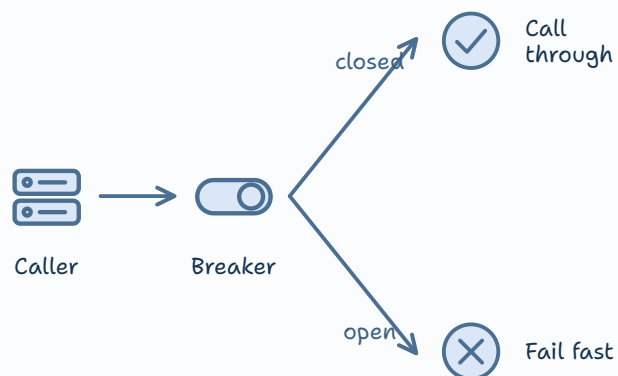
Split the write model from the read model so each is tuned for its job.

> RESILIENCE PATTERNS

SYSTEM DESIGN CONCEPTS

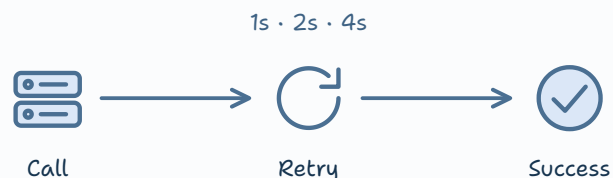


37 CIRCUIT BREAKER



Stop calling a failing dependency, fail fast, and probe until it recovers.

38 RETRY WITH BACKOFF



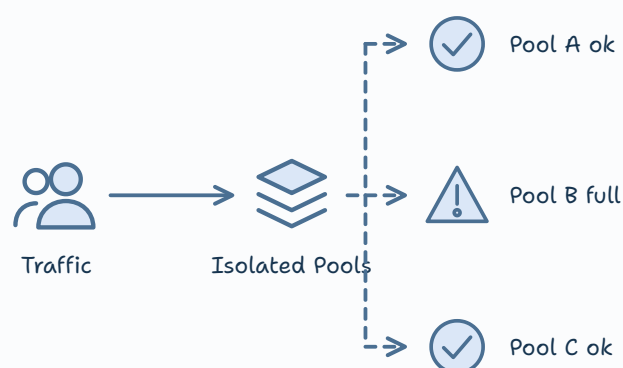
Retry transient failures on a growing, jittered delay — never in lockstep.

39 TIMEOUT



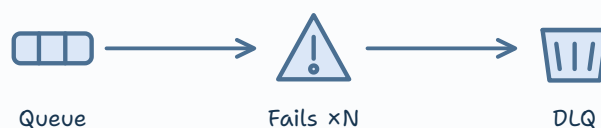
Every call gets a deadline; a hanging request is a failure that holds resources.

40 BULKHEAD



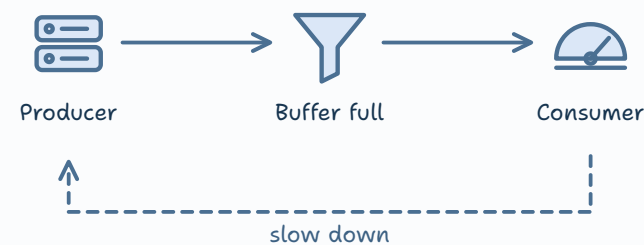
Give each dependency its own pool so one slow caller cannot starve the rest.

41 DEAD LETTER QUEUE



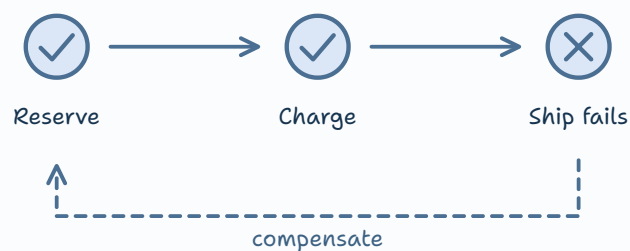
Park messages that keep failing instead of retrying them forever in place.

42 BACKPRESSURE



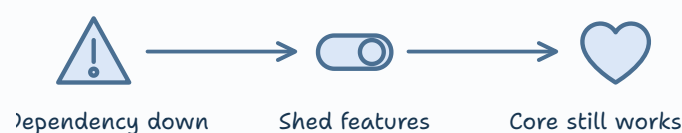
Signal upstream to slow down before the consumer's buffers overflow.

43 SAGA PATTERN



Model a long transaction as local steps, each with a compensating undo.

44 GRACEFUL DEGRADATION



Shed non-essential features under stress so the core product keeps serving.

45 CHAOS ENGINEERING

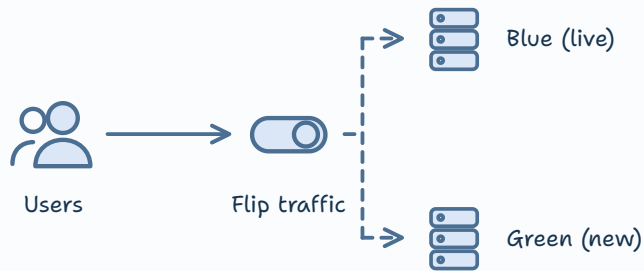


Inject failure on purpose in controlled experiments to find weaknesses first.

> SHIP & OPERATE

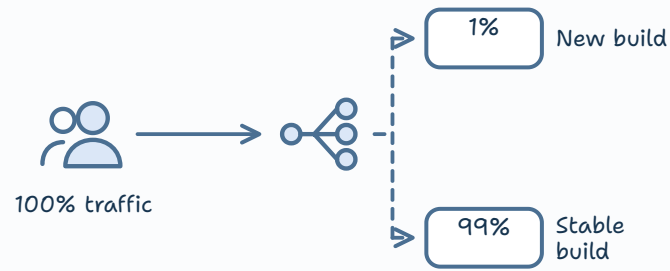
SYSTEM DESIGN CONCEPTS

46 BLUE-GREEN DEPLOY



Run two identical stacks and flip traffic between them — rollback is a flip back.

47 CANARY RELEASE



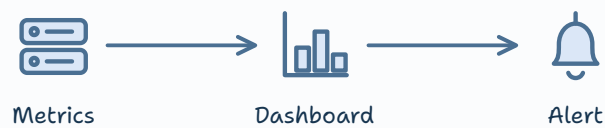
Send the new build to a small slice, watch the metrics, then widen or roll back.

48 FEATURE FLAGS



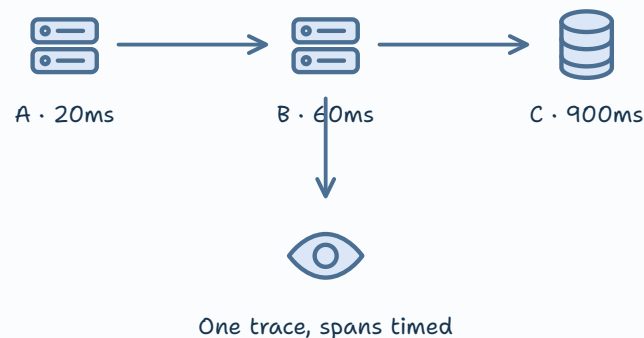
Turn behaviour on and off per cohort at runtime, with no deploy required.

49 MONITORING & ALERTING



Track metrics that describe user pain, and page a human only when one must act.

50 DISTRIBUTED TRACING



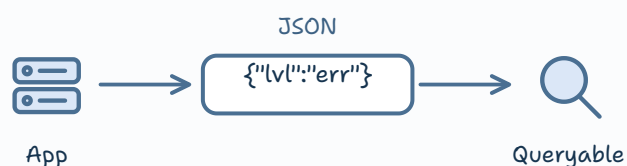
Follow one request across services and see exactly where the 900 ms went.

51 CORRELATION ID



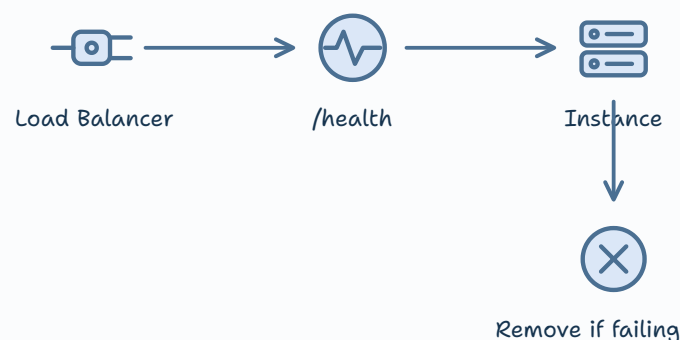
One ID travels with the request and stitches every log line back together.

52 STRUCTURED LOGGING



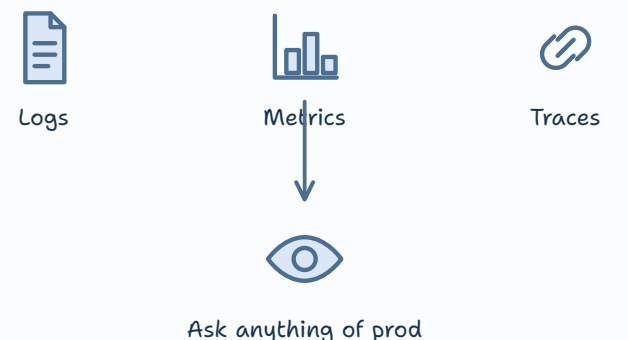
Emit machine-readable events, not prose, so logs can be filtered and aggregated.

53 HEALTH CHECKS



Instances report liveness and readiness; unhealthy ones leave the pool quietly.

54 OBSERVABILITY

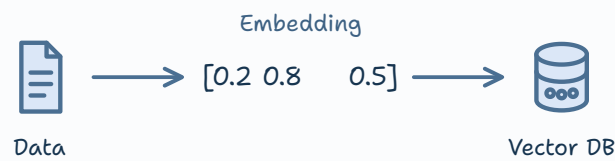


Logs, metrics and traces together let you ask new questions without a deploy.

> AI SYSTEM ARCHITECTURE

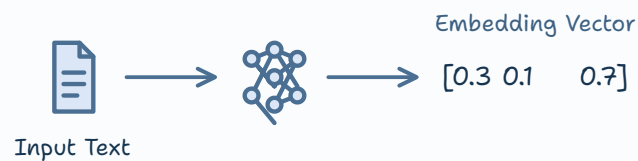
PART 1 OF 2

55 VECTOR DATABASE



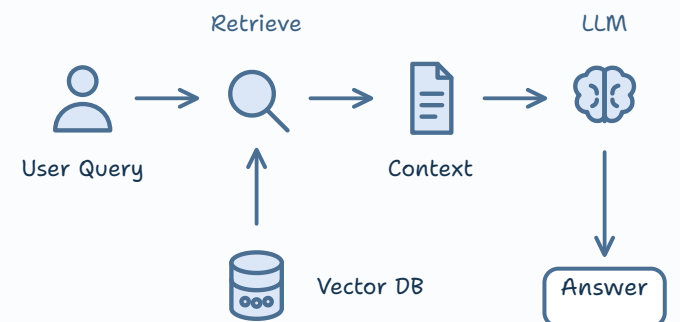
Stores embeddings as vectors for fast semantic search and retrieval.

56 EMBEDDING MODEL



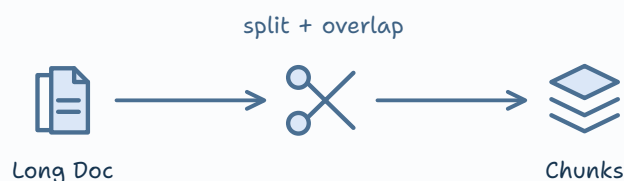
Converts text into high-dimensional vectors that capture semantic meaning.

57 RAG



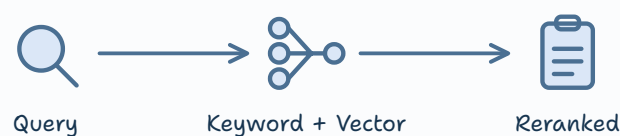
Retrieves relevant context from a knowledge base and augments the LLM response.

58 CHUNKING



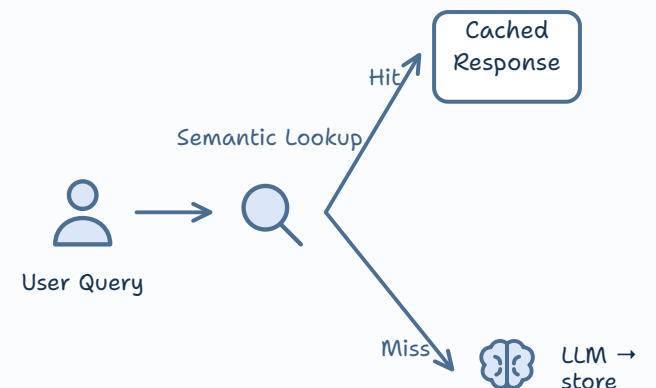
Split long documents into overlapping passages small enough to embed and retrieve.

59 HYBRID SEARCH



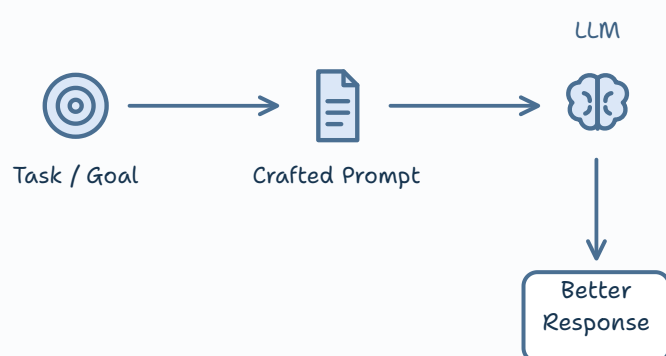
Blend keyword and vector results, then rerank so the best passages surface first.

60 SEMANTIC CACHE



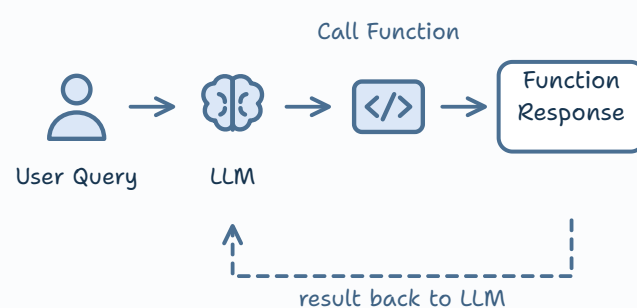
Caches similar queries and responses by meaning, not exact match, to cut cost.

61 PROMPT ENGINEERING



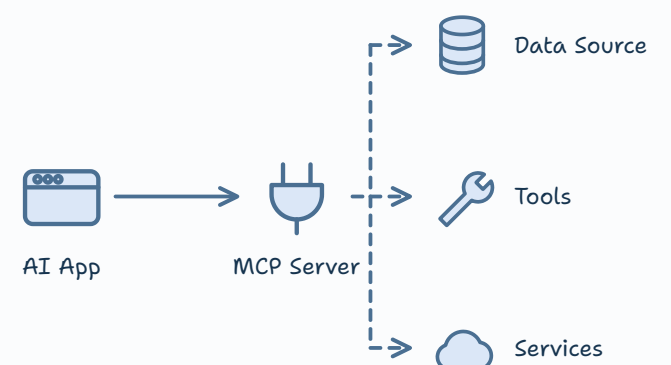
Designs effective prompts to guide LLMs toward accurate, relevant responses.

62 FUNCTION CALLING



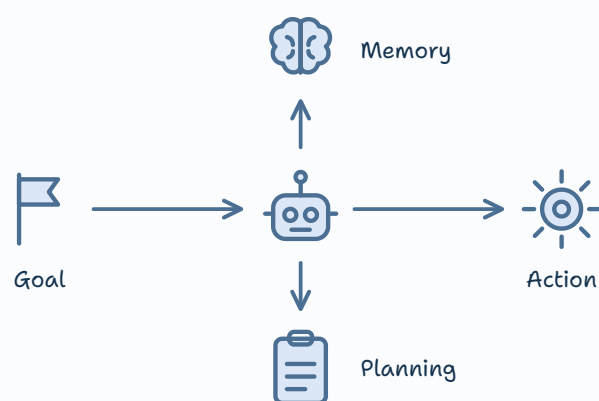
The LLM invokes external functions or APIs to fetch real-time data and act.

63 MCP



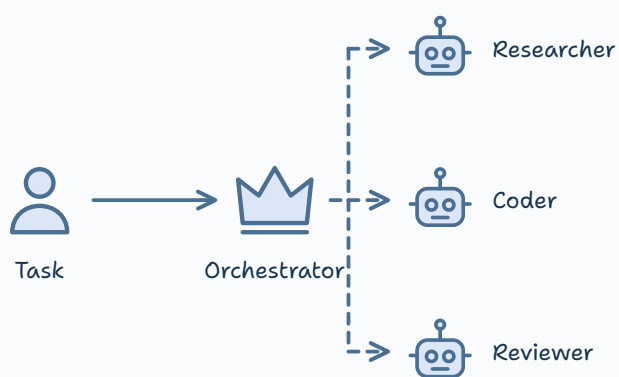
A standard protocol connecting models to external data, tools and services.

64 AI AGENT



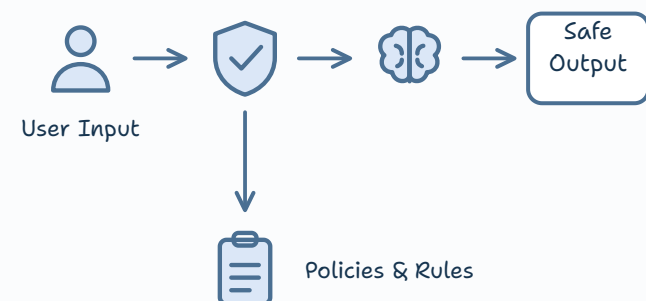
An autonomous system that perceives, plans and acts to achieve a goal.

65 MULTI-AGENT SYSTEMS



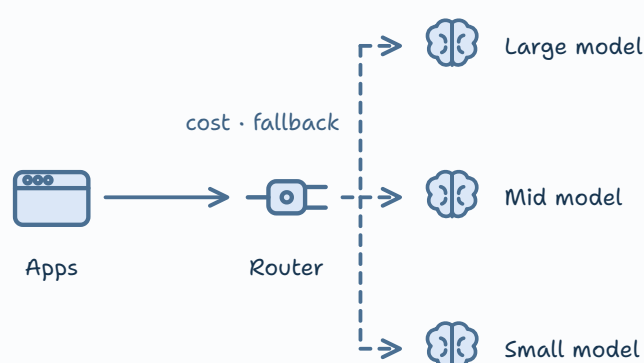
An orchestrator splits a task across specialist agents and merges their results.

66 GUARDRAILS



Enforces safety, policy and quality checks to keep AI output reliable.

67 LLM GATEWAY



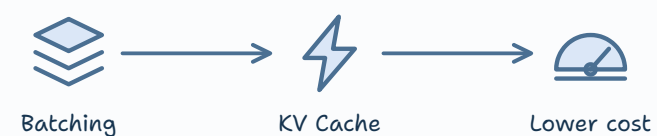
Routes requests across models by cost, latency and fallback — one API for all.

68 FINE-TUNING VS RAG



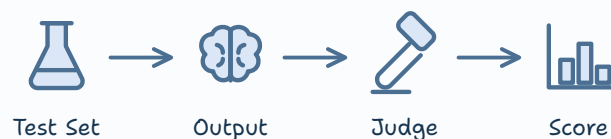
Fine-tuning teaches behaviour into weights; RAG supplies fresh facts at query time.

69 INFERENCE OPTIMISATION



Batching, KV caching and quantisation cut latency and cost per token served.

70 EVALS & LLM-AS-JUDGE



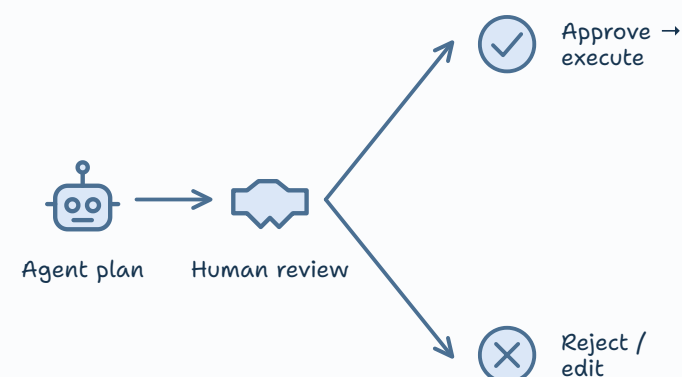
Score outputs against a fixed test set, often using a model as the grader.

71 LLM OBSERVABILITY



Trace prompts, tokens, cost and latency so regressions are visible, not guessed.

72 HUMAN IN THE LOOP



A person approves, edits or rejects high-stakes actions before they take effect.

> THANKS FOR READING >



Tauseef Fayyaz

Sharing insights on AI, Tech & Growth · Lead Full Stack Engineer · Building & scaling products

I build systems and figure out how developers grow — from a first line of code to shipping real products and scaling a career.

100K+
learners



BREAK IT DOWN

AI, coding & system design turned into practical insights



LEVEL YOU UP

Helping developers go from beginner → professional



BUILD FOR REAL

Scalable web & AI systems, shipped not theorised



MENTOR

Engineers coached into new jobs, switches and faster growth

IF THIS HELPED



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